Source-Seeded Four-Direction Horizon Refresh

Removing the Late Ambient Seed

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Abstract

The recursive packet chain of RH-93 removes ambient eigenspace resets inside one four-step block, but still starts that block from a late ambient leading packet. We move the only spectral seed to time zero and ask whether a low-dimensional Ritz chain can carry it through the complete frozen prefix.

If S is the source matrix, the initial normalized memory Gramian is

$$G_0 = \frac{S^*S}{\|S\|_F^2}.$$

Its leading rank-r packet is exactly the leading right singular subspace of S. Starting from this source packet, let V_{t-1} be the current packet, select k left singular directions of

$$K_t = (I - V_{t-1}V_{t-1}^*)G_tV_{t-1},$$

and retain the leading r Ritz directions in the resulting $(r+k)$-dimensional space. We prove a source-seeded recursive horizon theorem: every update preserves rank, improves the new-Gram predictor tail, and uses no ambient leading eigenspace after time zero. The exact gain remains certifiable by a generalized bottom frame of the compressed matrix.

A 384-bit audit follows ten channels from the source to the RH-93 endpoint, for a total of 120 primary updates. Width two has worst endpoint/reference tail ratio 11.397994, and width three has worst ratio 1.448940. Width four passes all ten endpoint gates, with worst ratio 1.001173, minimum projected-cross energy capture 97.536%, and compressed dimension at most 11. All 120 direct Ritz updates are monotone and all generalized frame gains are strictly positive. The source-SVD and initial-Gram projectors agree to at most 7.96×10^{-14} in operator norm.

Thus the late ambient seed can be removed on the archived horizons. The initial source-coordinate SVD, the ambient action G_tV_{t-1}, repeated all-level contraction, and continuum complexity bounds remain open. No Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis result is claimed.

Keywords: source Gramian; recursive Ritz method; projected cross operator; low-rank packet; validated numerics.

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1 Introduction

The late-memory route developed in the preceding layers seeks a finite-dimensional packet that captures the dynamically relevant part of a time-ordered Gramian. The rank clock of RH-82 suggests $r = O(\log(1/\sigma))$ at the archived scales [7]. RH-92 replaced a rigid pointwise contraction target by a four-step product budget [6]. RH-93 then showed that two projected-cross directions are enough to propagate the corrected packet recursively inside each selected four-step block [8].

That result still left one visible ambient operation. At the beginning of each late block, the seed packet was obtained from the leading eigenspace of the ambient Gramian. The reduced recursion began only after that reset. Such a finite construction is mathematically legitimate, but it cannot yet serve as an intrinsic all-prefix packet law.

The present paper asks the next minimal question:

> Can the packet be seeded once from the source at time zero and then carried recursively to the late endpoint without inserting any later ambient leading eigenspace?

There is a canonical seed. At time zero the state is the source S, so the normalized snapshot Gramian is $S^*S/\|S\|_F^2$. Its leading eigenspace is the leading right singular subspace of S. This observation is elementary, but it changes the architecture of the certificate: the seed is now attached to the input data rather than to a late dynamically assembled Gramian.

The algebraic part of the paper has two components. First, we record the source-seed equivalence with the correct spectral-gap qualification. Second, we formulate a recursive horizon theorem for an arbitrary finite sequence of positive semidefinite Gramians. It separates three operations:

1. apply the current Gramian to the incoming rank-r packet;
2. identify k projected-cross directions;
3. solve one $(r+k)$-dimensional Ritz problem.

No ambient leading eigenspace occurs after initialization. This is an exact finite-dimensional statement, independent of the numerical model.

The numerical question is not whether the recursion exists, but how wide the complement must be to retain endpoint accuracy over the complete prefix. The answer differs from the four-step contraction threshold. Two directions were enough locally in RH-93. Over the source-to-endpoint horizon, width two can lose more than a factor eleven relative to the ambient reference tail, and width three can lose a factor 1.45. Width four is the first tested width that is uniformly near the reference at all ten anchors.

This distinction is useful. A narrow packet can satisfy a selected contraction budget without identifying the leading packet accurately over a long prefix. Local contraction and horizon tracking are related but not identical route gates.

2 The source packet

Let $S : \mathbb{C}^m \to \mathbb{C}^n$ be nonzero. We regard its columns as source coordinates and define

$$G_0 = \frac{S^*S}{\|S\|_F^2}. \tag{1}$$

Then $G_0 = G_0^* \geq 0$ and $\operatorname{tr} G_0 = 1$.

Theorem 2.1 (Source-seed equivalence)**.** *Let*

$$S = U\Sigma Z^*$$

be a singular-value decomposition with singular values in nonincreasing order. For every rank r, the span of the first r columns of Z is a leading rank-r eigenspace of G_0. If $s_r(S) > s_{r+1}(S)$, this rank-r spectral subspace is unique. Without a gap, the two constructions determine the same family of admissible leading subspaces.

Proof. Equation (1) gives

$$G_0 = Z\frac{\Sigma^*\Sigma}{\|S\|_F^2}Z^*.$$

Thus the eigenvalues of G_0 are the squared singular values of S, divided by their sum, and the corresponding eigenvectors are the right singular vectors. A strict cutoff gap gives uniqueness of the spectral projector. When the cutoff is repeated, any rank-r choice within the tied eigenspace is simultaneously a leading right singular and a leading Gram eigenspace. □

Remark 2.2. The theorem does not eliminate a source-coordinate spectral operation. It identifies that operation intrinsically and places it at time zero. Replacing the source SVD by an analytic or structured seed construction is a later problem.

For the frozen model, let $X_t = T^t S$ and define the memory Gramians

$$G_t = \frac{X_t^* X_t}{\operatorname{tr}(X_t^* X_t)} + \eta G_{t-1}, \qquad \eta = \frac{1}{512}, \qquad t \geq 1. \tag{2}$$

The theorem applies because the recursion starts with (1). The particular formula (2) is used only in the audit; the recursive Ritz theorem below allows any finite positive semidefinite sequence.

3 Source-seeded recursive horizon theorem

Let $G_0, \dots, G_N$ be positive semidefinite matrices on a finite-dimensional Hilbert space $\mathcal{K}$. Let $V_0 : \mathbb{C}^r \to \mathcal{K}$ be an isometry spanning a leading rank-r eigenspace of G_0. Suppose V_{t-1} has already been constructed. Put

$$P_{t-1} = V_{t-1} V_{t-1}^*, \qquad K_t = (I - P_{t-1}) G_t V_{t-1}. \tag{3}$$

Choose an isometry $Q_t : \mathbb{C}^k \to \operatorname{Ran}(I - P_{t-1})$ spanning the first k left singular directions of K_t, omitting zero directions if the cross rank is smaller. Set $Z_t = [V_{t-1}, Q_t]$ and

$$H_t = Z_t^* G_t Z_t = \begin{pmatrix} A_t & B_t \\ B_t^* & D_t \end{pmatrix}. \tag{4}$$

Let C_t contain the leading r orthonormal eigenvectors of H_t and define

$$V_t = Z_t C_t. \tag{5}$$

For an isometric packet V, write

$$\mathcal{E}_G(V) = \operatorname{tr} G - \operatorname{tr}(V^* G V).$$

Theorem 3.1 (Source-seeded recursive horizon theorem)**.** *The recursion* (3)–(5) *has the following properties.*

1. *Every V_t is an isometry of rank r and is constructed in a space of dimension at most $r + k$.*
2. *At every update,*
$$\mathcal{E}_{G_t}(V_t) \leq \mathcal{E}_{G_t}(V_{t-1}). \tag{6}$$
3. *If the eigenvalues of H_t are written in increasing order, the exact captured-energy gain is*
$$\Delta_t = \operatorname{tr} D_t - \sum_{j=1}^{k} \lambda_j^{\uparrow}(H_t) \geq 0. \tag{7}$$
4. *After V_0 is chosen, no leading eigenspace of an ambient G_t is needed anywhere in the recursion.*

Proof. The columns of Z_t are orthonormal, so (5) is an isometry with r columns. Its range lies in $\operatorname{Ran} Z_t$, whose dimension is at most $r + k$.

The old coordinate packet $V_{t-1} = Z_t[I_r, 0]^T$ is an admissible rank-r trial subspace in $\operatorname{Ran} Z_t$. Ky Fan's maximum principle therefore implies that the leading rank-r Ritz packet captures at least as much G_t-energy as V_{t-1}, proving (6) [1, 2]. The sum of the leading r eigenvalues of H_t is

$$\operatorname{tr} H_t - \sum_{j=1}^{k} \lambda_j^{\uparrow}(H_t).$$

Subtracting the old capture $\operatorname{tr} A_t$ and using $\operatorname{tr} H_t = \operatorname{tr} A_t + \operatorname{tr} D_t$ gives (7). Finally, (3) requires $G_t V_{t-1}$, and (5) requires only the compressed eigensolve for H_t. Neither operation asks for a leading eigenspace of G_t. □

Remark 3.2 (Meaning of ambient-free seed). The late ambient *eigenspace* seed is removed. The matrix action $G_t V_{t-1}$ remains ambient. This distinction is essential: the theorem is a low-dimensional spectral reduction, not yet a continuum complexity result.

3.1 Projected-cross selection

The projected cross operator has only r columns. Its selected directions maximize captured cross Frobenius energy:

$$\max_{\substack{Q^*Q=I_k \\ Q^*V=0}} \|Q^*GV\|_{S_2}^2 = \sum_{j=1}^{k} s_j\big((I - VV^*)GV\big)^2. \tag{8}$$

This is the Ky Fan principle applied to KK^*, or equivalently the Eckart–Young singular-subspace characterization [3]. The unselected cross energy is

$$\|(I - QQ^*)K\|_{S_2}^2 = \sum_{j>k} s_j(K)^2. \tag{9}$$

One useful reduced identity is already visible:

$$K^*K = V^*G(I - VV^*)GV = V^*G^2V - (V^*GV)^2. \tag{10}$$

Thus the cross singular values and right singular vectors are determined by an $r \times r$ positive semidefinite matrix. Turning (10) into a complete stable factorization, including small-singular-value thresholds and reconstruction of the left directions, is the natural next layer.

3.2 Generalized bottom-frame certificate

For a full-column-rank frame $W \in \mathbb{C}^{(r+k)\times k}$ define

$$\mathcal{R}_{H_t}(W) = \operatorname{tr}\big((W^*W)^{-1}W^*H_tW\big).$$

Ky Fan's minimum principle gives

$$\sum_{j=1}^{k} \lambda_j^{\uparrow}(H_t) \le \mathcal{R}_{H_t}(W).$$

Consequently

$$\operatorname{tr} D_t - \mathcal{R}_{H_t}(W) > 0 \quad \Longrightarrow \quad \Delta_t > 0. \tag{11}$$

The generalized form is invariant under a nonsingular change of frame coordinates and is therefore appropriate for outward-rounded verification.

Table 1: Outward upper bounds for endpoint tail divided by the ambient leading-packet tail. Width four passes the 1.01 gate in all channels.

channel	width 2	width 3	width 4
0.16 left	1.000004	1.000001	1.000001
0.16 right	1.232685	1.000006	1.000007
0.08 left	1.163776	1.000500	1.000001
0.08 right	11.397994	1.021816	1.000001
0.04 left	2.778159	1.004185	1.000001
0.04 right	6.891207	1.448940	1.000001
0.02 left	1.033230	1.003791	1.000003
0.02 right	3.623212	1.143370	1.000002
0.01 left	1.025773	1.000862	1.000006
0.01 right	5.939843	1.014986	1.001173

4 Validated experiment

4.1 Frozen protocol

We use the same five noise scales, two directional channels, normalized memory parameter, and half-log rank clock as RH-82–RH-93. The scales are

$$\sigma \in \{0.16, 0.08, 0.04, 0.02, 0.01\}.$$

For inherited horizons $4, 9, 16, 25, 32$, respectively, the late endpoints are

$$N_\sigma = \max\{4, \lceil 2H_\sigma/3 \rceil\} \in \{4, 6, 11, 17, 22\}.$$

Each channel begins from the leading right singular packet of its source. The complete chain from $t = 1$ through N_σ is then run independently at widths two, three, and four.

All source matrices, Gramians, packets, compressed frames, and reported tail forms are interpreted as exact binary inputs to Arb at 384-bit precision. Outward endpoints are used for every inequality. The SVD, QR factorization, and symmetric eigensolves select trial subspaces in double precision; Arb then certifies the claimed quadratic forms for those concrete subspaces. This is a standard validated-numerics division of labor [4, 5].

The primary width-four audit contains

$$2(4 + 6 + 11 + 17 + 22) = 120$$

updates. At every update we record the predictor tail, corrected tail, ambient reference tail, generalized frame gain, frame metric determinant, cross-energy fraction, and orthogonality defects.

4.2 Endpoint width threshold

Table 1 gives the central finite result. Width two is not a robust horizon-tracking law, despite its success on the selected late four-step contraction windows. Width three substantially improves the chain, but one channel still reaches 1.448940. Width four remains within 0.118% of the reference in every channel.

This is not a theorem that four directions are universally minimal. It is a sharp threshold among the three archived constructions at the fixed rank clock and frozen horizons. In particular, recursive chains of different widths diverge after their first update, so one cannot compare their endpoints by a simple nesting argument.

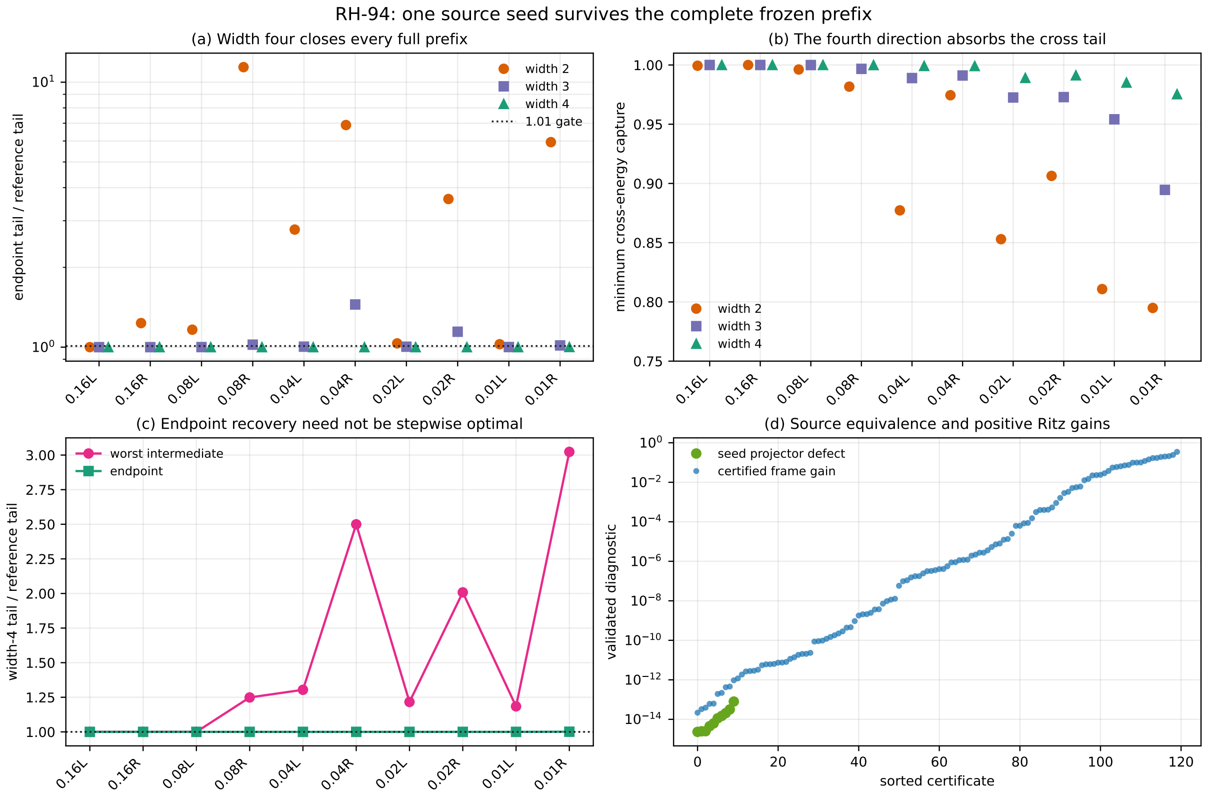

Figure 1: Full-prefix source-seeded refresh. Width four closes all endpoint gates, captures at least 97.536% of projected-cross energy, and recovers at the endpoint even when intermediate reference ratios are larger.

4.3 Mechanism diagnostics

The minimum selected projected-cross energy fractions over all updates are

$$0.7949096 \quad (k=2), \qquad 0.8945814 \quad (k=3), \qquad 0.9753653 \quad (k=4).$$

The fourth direction therefore removes most of the residual cross tail left by width three. This diagnostic is consistent with the endpoint transition, although (9) alone does not control the Ritz tail: the complement diagonal block and joint rotation also matter.

All 120 width-four corrected tails are no larger than their incoming predictor tails. Every generalized frame gain in (11) is strictly positive; the smallest validated lower bound is 2.16×10^{-14}. The smallest frame metric determinant is larger than 0.999999999999996, and the largest orthogonality defect is below 2.74×10^{-15}. The compressed dimension is at most $r+4=11$.

The source seed was also computed independently from the source SVD and from the leading eigenspace of G_0. The largest operator-norm distance between the two numerical projectors is 7.96×10^{-14}, in agreement with Theorem 2.1.

The endpoint result should not be confused with pointwise reference tracking. The largest intermediate width-four tail/reference ratio is 3.023179. The recursive packet may temporarily depart from the ambient leading packet and recover later. The theorem guarantees predictor improvement, while the audit certifies endpoint recovery.

5 Consequences for the route

RH-94 changes the finite architecture in one precise way. A late ambient leading eigenspace is no longer needed to start the reduced recursion. The entire frozen prefix can be represented as

$$\text{source packet} \longrightarrow \text{projected-cross action} \longrightarrow \text{small Ritz refresh} \longrightarrow \cdots \longrightarrow \text{late endpoint.}$$

It is useful to keep two width statements separate:

- *two-direction block law*: sufficient for the RH-93 four-step contraction target on the selected late windows;
- *four-direction horizon law*: sufficient for near-reference source-to-endpoint tracking on all ten frozen channels.

The first is economical for contraction. The second is stronger for packet identification. A future proof may use the width-four chain during burn-in and switch to width two inside a stable repeated-block regime.

The next obstruction is also clearer. Equation (10) reduces direction *selection* to an $r \times r$ matrix, but constructing the left directions still uses $G_t V_{t-1}$. A reduced factorization should:

1. diagonalize $K_t^* K_t$ rather than an ambient cross matrix;
2. reconstruct only directions whose singular values exceed a certified threshold;
3. quantify how discarded cross singular values perturb the corrected tail;
4. state exactly which moments $V^* G^j V$ and ambient packet actions remain.

That is a bounded and testable next problem, rather than an appeal to an unspecified continuum limit.

6 Claim boundary

The present result is deliberately finite.

1. The source SVD is an initial source-coordinate spectral operation. It has been identified, not eliminated.
2. Applying G_t to a rank-r packet remains an ambient operation. The spectral solve is reduced, but a continuum complexity theorem is absent.
3. Only five frozen scales and two channels are audited. No uniform all-level width-four theorem follows from ten anchors.
4. Endpoint recovery does not imply stepwise closeness to the ambient leading packet.
5. No repeated-block contraction theorem, normalization/observability bridge, moving-cloud trace-class limit, Hilbert–Polya operator, zeta-zero identification, or proof of the Riemann Hypothesis is obtained.

7 Conclusion

The late ambient seed in the recursive packet route is not structurally necessary on the archived horizons. The leading source right singular subspace is exactly the initial Gram packet, and four projected-cross directions carry that packet through all 120 updates with near-reference endpoint tails. Widths two and three do not provide the same robust horizon tracking, so the full-prefix problem reveals a new finite width threshold not visible in the late four-step contraction test.

The route now begins at a genuine source object and remains recursively low-dimensional in its spectral solves. Its next mathematical task is the projected-cross Gram reduction (10): turn the ambient cross SVD into a stable small-matrix factorization and derive a quantitative tail criterion for discarded directions.

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